

HWRS 561b: Physical Hydrogeology II

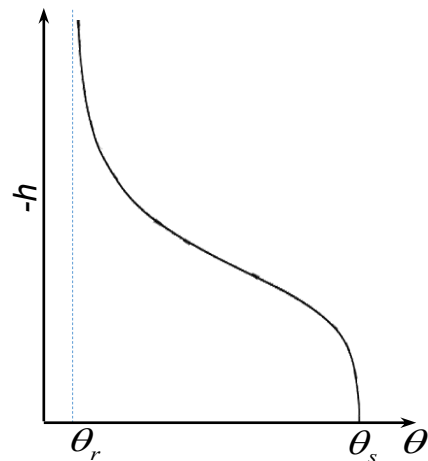
PFAS (part 2)

Modeling PFAS Transport in the Vadose Zone

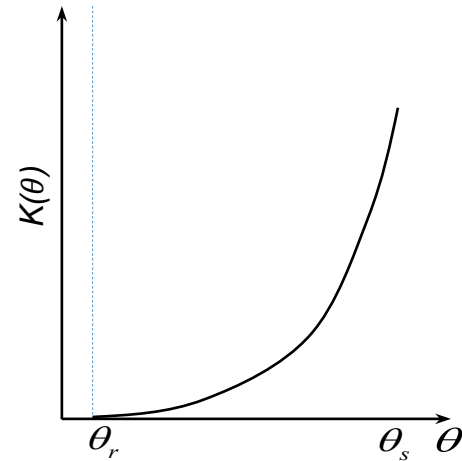
HWRS 561b
Bo Guo
Spring 2026

Variably saturated flow in the vadose zone

$$\frac{\partial \theta}{\partial t} - \nabla \cdot [\mathbf{K}(\theta) \nabla (h - z)] = 0$$



SWC

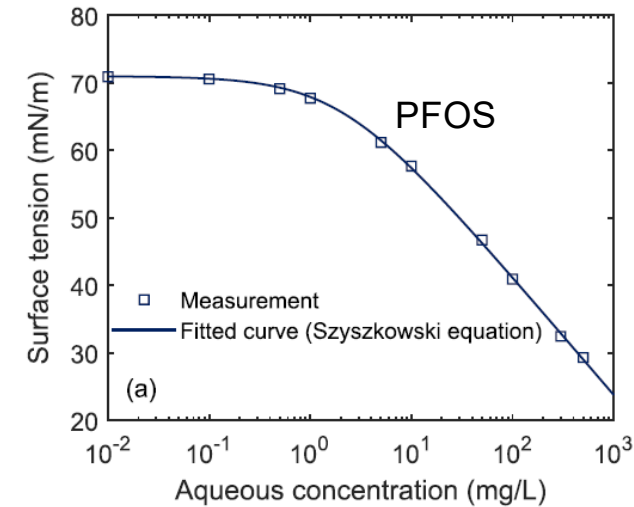


Unsat hydraulic conductivity

PFAS dissolving in water changes surface tension

$$\sigma = \sigma_0 \left[1 - b \ln \left(1 + \frac{C}{a} \right) \right]$$

Annotations: σ is Surface tension of water w/o PFAS; σ_0 is Surface tension of water w/o PFAS; b and a are Fitting parameters; C is PFAS concentration.



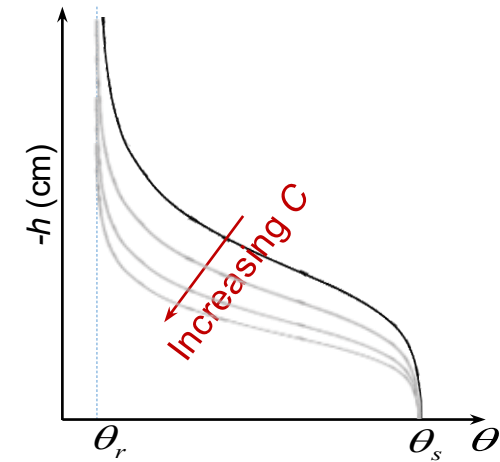
Question: Will the SWC and $K - \theta$ change in the presence of PFAS? If so, how?

Modeling PFAS Transport in the Vadose Zone

Let's employ the capillary tube model to think about the problem (Leverett-J scaling)

w/o PFAS	w/ PFAS
$-h_0 = \frac{2\sigma_0 \cos \gamma_0}{\rho g r}$	$-h = \frac{2\sigma \cos \gamma}{\rho g r}$
$\Rightarrow h = \frac{\sigma \cos \gamma}{\sigma_0 \cos \gamma_0} h_0$	$\Rightarrow \theta = \begin{cases} \theta_r + \frac{\theta_s - \theta_r}{\left[1 + \left(\frac{\sigma_0 \cos \gamma_0}{\sigma \cos \gamma} \alpha h \right)^n \right]^m}, & h < 0 \\ \theta_s, & h \geq 0 \end{cases}$

Scaling factor



Insights:

(1) We can think of this as PFAS modifies the α in the V-G model.

$$\alpha_{\text{modified}} = \frac{\sigma_0 \cos \gamma_0}{\sigma \cos \gamma} \alpha$$

(2) Now, the variably saturated flow depends on PFAS transport.

Modeling PFAS Transport in the Vadose Zone

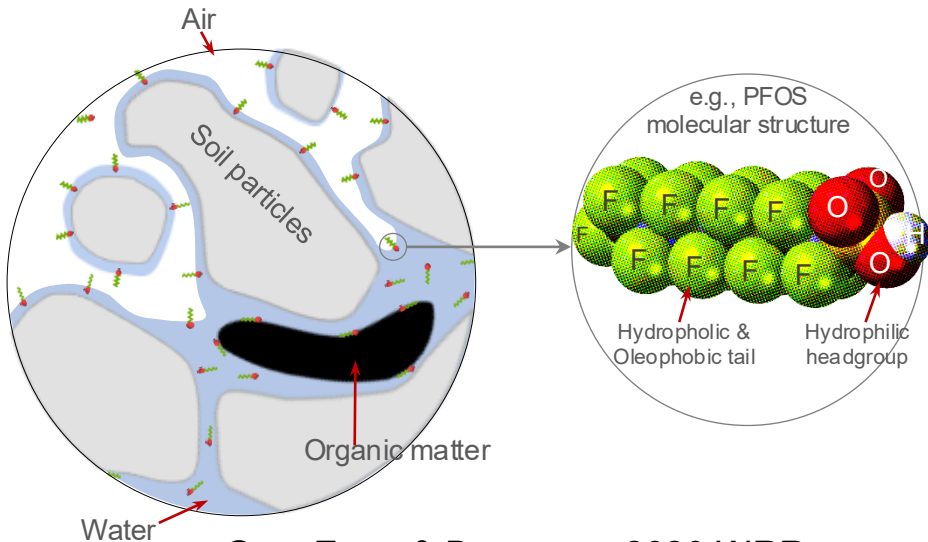
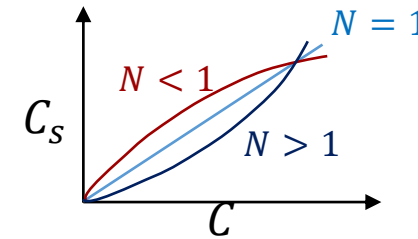
PFAS transport in variably unsaturated flow (neglecting the interactions among PFAS)

$$\frac{\partial(\theta C)}{\partial t} + \rho_b \frac{\partial C_s}{\partial t} + \frac{\partial C_{aw}}{\partial t} + \nabla \cdot (\theta \mathbf{v} C) - \nabla \cdot (\theta \mathbf{D} \nabla C) = 0$$

Storage in the
aqueous phase

Storage in the
solid-phase
adsorption

Storage in the air-water
interfacial adsorption



Solid-phase
adsorption

$$C_s = K_f C^N$$

Air-water
interfacial
adsorption

$$C_{aw} = \Gamma A_{aw} = K_{aw} C A_{aw} \quad \sigma = \sigma(C) \quad \text{Szyszkowski equation}$$

$$K_{aw} = \frac{\Gamma}{C} = -\frac{1}{RTC} \left(\frac{\partial \sigma}{\partial \ln C} \right)_T$$

$$\Rightarrow K_{aw} = \frac{1}{RT} \frac{\sigma_0 b}{a + C}$$

Modeling PFAS Transport in the Vadose Zone

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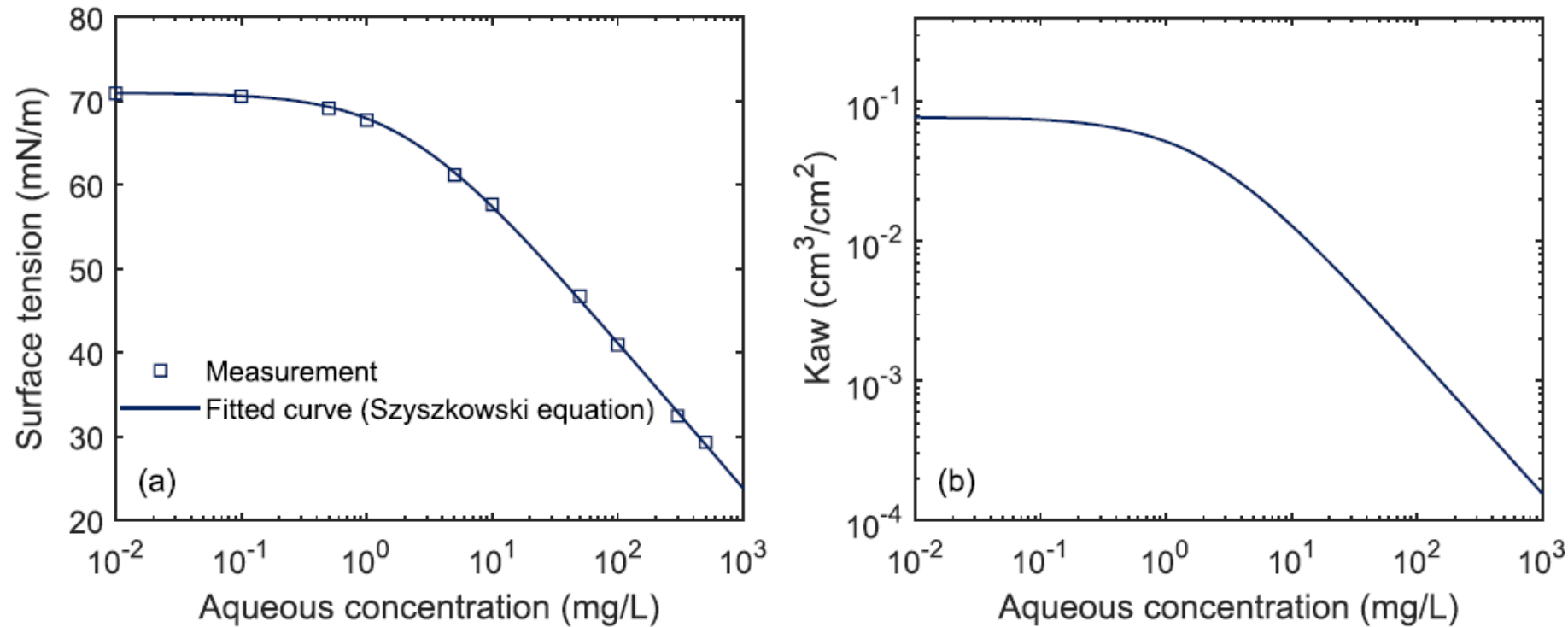


Figure 3. (a) Measured surface tension as a function of the aqueous concentration of PFOS. The solid line is a fitted curve using the Szyszkowski equation. (b) The computed air-water interfacial adsorption coefficient (K_{aw}) as a function of the aqueous concentration of PFOS.

Guo, Zeng & Brusseau. 2020 WRR

Simplified Models

- Steady-state unsaturated flow:

$$q = K$$

- PFAS transport:

$$\frac{\partial(\theta C)}{\partial t} + \boxed{\rho_b \frac{\partial C_s}{\partial t}} + \boxed{\frac{\partial C_{aw}}{\partial t}} + \underbrace{\nabla \cdot (\theta \mathbf{v} C)}_{\text{Advection}} - \underbrace{\nabla \cdot (\theta \mathbf{D} \nabla C)}_{\text{Dispersion}} = 0$$

All the equations are linearized and can be solved analytically (Guo et al., 2022). The solutions are in the forms of integrals, which can be implemented in Excel spreadsheet (See additional details in the PFAS-LEACH Excel User Guide (Ma et al. 2025)).

Linear and rate-limited solid-phase adsorption

$$C_{s,1} = F_s K_d C \quad \text{Instantaneous}$$
$$\frac{dC_{s,2}}{dt} = \alpha_s [(1 - F_s) K_d C - C_{s,2}] \quad \text{Kinetic}$$

Linear and equilibrium air-water interfacial adsorption

$$C_{aw} = A_{aw} K_{aw} C \quad \text{Instantaneous}$$

K_{aw} can be assumed as a constant at lower concentrations, which is often the case at PFAS-contaminated sites

